Online Supplement

Synthetic Cultural Agents as Group-Level Instruments: A Preregistered Evaluation from Anchor to Respondents

This supplement reports the two registered descriptives that the article sends here, the contrasts with the questionnaire readers and the prompt route, the factorial own effects with their t statistics, the first-cohort three-link table, the full tables behind several sentences of the article, the detail of the descriptive follow-up analyses, the items the probe selected, the provenance of the scoring runs, and the design detail and robustness numbers that the article summarizes. Every statistic is read from the same numbers table as the article and keeps the article's surface labels (S1 and S2 for the first and second cohorts on the strict battery, H for the retrospective held-out screen, R for the unseeded retrain, N for the near-zero refits, and D for the descriptive follow-up analyses, which were not preregistered). Sections and tables numbered with an S belong to this supplement. As in the article, every result on surface H is retrospective, because the WVS Wave 7 file existed before the probe and is the only human source.

S1 Order of work for the held-out screen

Figure S1 gives the order of work of the held-out screen described in the registration section of the article.

S2 Registered descriptives

Item-level agreement without the smallest items (H). Section 8 of the registration lists one item-level descriptive beside the primary statistic: the selected items whose signed agreement satisfies $|a_i| \geq 0.15$. The registration excludes this cut from the inference, because it conditions on the human criterion. Of the 56 items selected under the primary rule, 26 reach the cut. Their mean signed agreement is 0.115, and 69.2% of them have $a_i > 0$, against 60.7% over all 56 selected items. That is, the selected items whose human country differences correlate clearly with the GPS coordinate of their assigned dimension run in the assigned direction about two times

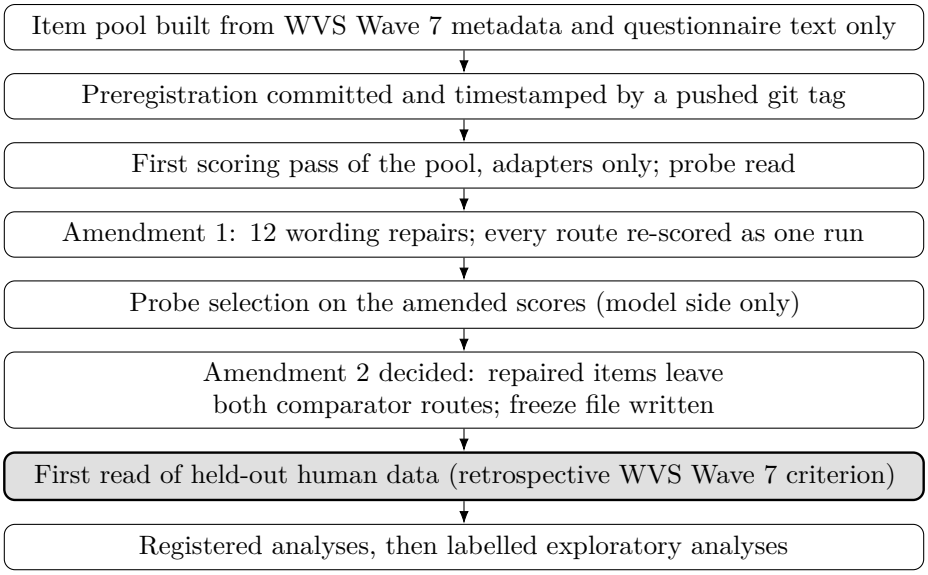


Figure S1: Order of work for the held-out screen. Every step above the shaded box used model scores, metadata or questionnaire text only. The shaded box is the single read of held-out human data.

in three. The figure describes where the agreement in A comes from and carries no interval. It is no evidence for the screen beyond the registered test reported in the article.

The face-valid reader on all of its items (H). Section 6 of the registration reports the face-valid reader's statistic on all of its assigned items, negative reciprocity included, beside the registered contrast on the imprint set. The registration counted 30 assigned items. Amendment 2 removed the one repaired item the reader had assigned (Q82), which leaves 29. Over these items the reader's A is 0.055, against 0.095 on its 25 imprint-set items. The difference comes from its 4 negative-reciprocity items, whose A is -0.192: the human country differences on the items the reader linked to negative reciprocity by their wording run, on average, against the direction it read from their response labels. These items lie outside the imprint set and enter none of the registered contrasts.

S3 Per-dimension readings of the questionnaire readers

The registration asks for per-dimension A_d of both questionnaire readers beside the coverage-matched contrast (Table S1). The intervals come from the same registered country bootstrap as the adapters' per-dimension readings in the article, and the

repaired items are removed as Amendment 2 requires. Every interval covers zero, and most cells hold a handful of items, so the table shows mainly where each reader concentrates. Both readers assign most of their imprint-set items to trust, which is why their own-item A in the article is close to their trust reading.

Table S1: Per-dimension readings of the two questionnaire readers on their own imprint-set items (H, retrospective).

	Face-valid reader		Typed decision model	
Dimension	A_d [95% interval]	Items	A_d [95% interval]	Items
Trust	0.110 [−0.154, 0.379]	19	0.151 [−0.076, 0.370]	32
Risk taking	0.044 [−0.170, 0.243]	4	−0.074 [−0.398, 0.239]	1
Altruism	0.055 [−0.163, 0.265]	2	0.049 [−0.078, 0.165]	7

Note: The typed decision model's assignment is the majority over 5 runs. Neither reader assigned an item to positive reciprocity. Intervals are percentile limits from 2,000 resamples of the 42 countries (seed 20261015), unadjusted for the number of dimensions.

S4 Contrasts with the questionnaire readers and the prompt route

Table S2 gives every registered contrast of the held-out screen that the article summarizes in its screen figure: the adapters against the face-valid reader, the typed decision model and the sign-profile prompt, on the items both routes assign and on each route's own items, with the number of shared items and the number on which the two routes agree in dimension and sign. Every interval covers zero. The coverage-matched contrasts rest on few items, and an interval that covers zero on so few items is consistent with a match and does not establish one.

S5 The second scenario bank in full

The article reports the adapter–GPS correlations of all seven strict-battery surfaces on the second bank, and those of the four imprint surfaces on the first. Table S3 adds the first-cohort values of the three flat surfaces, so that both cohorts are shown on every surface. Both cohorts use the same statistic, the same profile-block bootstrap and the same seed. Patience and negative reciprocity are flat on both banks, and their intervals are wide on both.

The article reports the bank-to-bank Pearson correlation for five surfaces. Table S4 gives both the Pearson and the Spearman coefficient for all seven, over the 26 profiles the two banks share. The two coefficients tell the same story. Risk taking is the

Table S2: Retrospective screen: the adapters against two questionnaire readers and against prompting with the same signs.

	Face-valid reader	Typed decision model, majority of 5 runs	Sign-profile prompt
Items assigned by both routes	5	9	16
Same dimension and sign on those items	1	7	6
Adapter minus route, shared items	0.020 [−0.181, 0.215]	−0.002 [−0.064, 0.061]	0.004 [−0.065, 0.075]
Adapter minus route, own items	−0.034 [−0.242, 0.169]	−0.067 [−0.237, 0.103]	0.033 [−0.083, 0.143]
Route's own A (items)	0.095 (25)	0.128 (40)	0.028 (56)

Note: Differences are adapter minus the named route, with 95% intervals from the paired country bootstrap (2,000 resamples). The registered primary contrast with each questionnaire reader is the one on shared items. The questionnaire readers' own items are restricted to the imprint set, with the 12 items repaired by Amendment 1 removed. The adapter's own A is 0.061 on 56 items. Every interval in the table covers zero.

Table S3: Adapter–GPS Spearman correlation over the 42 countries, by surface and cohort (S1 and S2, model side only).

Surface	First cohort, original bank (S1)	Second cohort, second bank (S2)
Trust	0.744 [0.500, 0.865]	0.803 [0.615, 0.876]
Risk taking	0.435 [0.085, 0.686]	0.210 [−0.199, 0.584]
Positive reciprocity	0.488 [0.120, 0.695]	0.366 [−0.068, 0.652]
Altruism	0.406 [0.020, 0.676]	0.364 [−0.057, 0.653]
Negative reciprocity	−0.028 [−0.418, 0.406]	0.239 [−0.237, 0.604]
Patience (Q43)	−0.019 [−0.414, 0.452]	0.005 [−0.403, 0.470]
Patience (Q50)	0.177 [−0.313, 0.580]	0.161 [−0.343, 0.557]

Note: Percentile intervals from a profile-block bootstrap over the 26 sign profiles the countries realize (2,000 draws, seed 20261011). In the second cohort each country takes the adapter of its sign profile.

least stable surface across banks on either coefficient, and the two patience items, flat against the GPS, are nevertheless stable across banks. That is, a surface can be a stable property of the instrument and still carry no agreement with its anchor.

Table S4: Agreement between first-bank and second-bank surface composites over the shared sign profiles (model side only).

Surface	Pearson [95% interval]	Spearman [95% interval]
Trust	0.909 [0.797, 0.965]	0.907 [0.774, 0.957]
Risk taking	0.598 [0.229, 0.819]	0.544 [0.187, 0.772]
Positive reciprocity	0.898 [0.775, 0.960]	0.937 [0.818, 0.983]
Altruism	0.949 [0.859, 0.989]	0.944 [0.832, 0.985]
Negative reciprocity	0.708 [0.488, 0.861]	0.600 [0.230, 0.838]
Patience (Q43)	0.812 [0.699, 0.945]	0.889 [0.682, 0.971]
Patience (Q50)	0.940 [0.890, 0.975]	0.953 [0.856, 0.982]

Note: First-bank values are the means of the first-cohort country adapters within each profile, and second-bank values are the profile adapters' composites. Percentile intervals from a 2,000-draw bootstrap over profiles. No human data enter.

S6 The country-named persona route

The persona route replaces the neutral opening of the evaluation prompt with a sentence that names the country, and it is scored on the base model with no adapter. One run scored all 42 personas after a fresh base-model pass had reproduced the banked base-model scores of the second cohort's strict run, with a largest log-probability gap of 0.0 over 140 option rows. Table S5 gives the GPS correlation of the two prompt routes by surface; the adapter columns are those of Table S3. Table S6 gives each route's correlation minus the persona route's. The planned contrast is the first-cohort difference on trust, and every other difference is descriptive. Under a country bootstrap (seed 20260905) the planned contrast is 0.716 [0.325, 1.075].

The persona route tracks none of the four imprint surfaces. Its two clear correlations are on the two patience items, and they have opposite signs.

On the 16 countries of the development stage, rescored on the frozen stack with the first-cohort adapters, the trust correlation is 0.818 [0.554, 0.926] for the adapters and −0.038 [−0.565, 0.480] for the persona route, with a country bootstrap. These values are descriptive. The development stage used other adapters on an earlier software stack, and its own scores enter no statistic in the article.

Table S5: Prompt routes against the GPS: Spearman correlation over the 42 countries, by surface (model side only).

Surface	Sign-profile prompt	Persona (profile blocks)	Persona (country bootstrap)
Trust	0.773 [0.584, 0.844]	0.028 [−0.339, 0.381]	[−0.273, 0.330]
Risk taking	−0.448 [−0.722, −0.049]	0.195 [−0.201, 0.545]	[−0.151, 0.506]
Positive reciprocity	0.490 [0.150, 0.709]	0.033 [−0.342, 0.315]	[−0.245, 0.320]
Altruism	0.540 [0.199, 0.747]	−0.136 [−0.485, 0.214]	[−0.427, 0.197]
Negative reciprocity	0.005 [−0.440, 0.447]	0.157 [−0.140, 0.449]	[−0.153, 0.458]
Patience (Q43)	−0.195 [−0.546, 0.218]	−0.491 [−0.771, −0.124]	[−0.721, −0.214]
Patience (Q50)	0.184 [−0.299, 0.573]	0.509 [0.157, 0.730]	[0.243, 0.714]

Note: The sign-profile prompt column maps each country to its prompted profile. Profile-block intervals resample the 26 sign profiles (2,000 draws, seed 20261011); the last column resamples countries (seed 20260905) and repeats the point estimate of the column before it.

Table S6: Each route's GPS correlation minus the persona route's, by surface (model side only).

Surface	First cohort − persona	Second cohort − persona	Sign prompt − persona
Trust*	0.716 [0.191, 1.154]	0.775 [0.347, 1.144]	0.745 [0.327, 1.086]
Risk taking	0.240 [−0.156, 0.617]	0.015 [−0.436, 0.493]	−0.643 [−1.060, −0.118]
Positive reciprocity	0.454 [−0.036, 0.907]	0.332 [−0.196, 0.816]	0.457 [0.052, 0.884]
Altruism	0.543 [0.042, 0.990]	0.500 [−0.026, 0.965]	0.677 [0.198, 1.083]
Negative reciprocity	−0.185 [−0.612, 0.297]	0.082 [−0.441, 0.561]	−0.152 [−0.640, 0.374]
Patience (Q43)	0.472 [0.019, 0.986]	0.496 [0.031, 1.013]	0.295 [−0.134, 0.720]
Patience (Q50)	−0.333 [−0.840, 0.085]	−0.349 [−0.875, 0.084]	−0.326 [−0.802, 0.083]

Note: *The planned contrast, fixed before any persona was scored; every other cell is descriptive. Both correlations of a difference are recomputed on every draw of one profile-block bootstrap (2,000 draws, seed 20261011).

S7 The sign-profile factorial in full

Table S7 gives, for each of the seven strict-battery surfaces, the main effect of the surface's own sign bit and its t statistic on the trained route and on the sign-profile prompt, with the summary statistics of entanglement. The article's factorial figure draws the same own effects without an interval, because the table banks none; the t statistics here carry the uncertainty of each own effect (surface F; model side only). Cross effects between bits and surfaces that the article cites are given in Section S13.

Table S7: Own effects of the six sign bits and summary of entanglement, by route. Strict battery, all 64 sign profiles.

	Trained adapters		Sign-profile prompt	
Own effect on surface	Main effect	t	Main effect	t
Trust	0.364	16.74	1.780	394.46
Negative reciprocity	0.625	11.64	−0.023	−1.80
Altruism	0.242	6.32	0.364	11.99
Patience (Q50)	0.133	6.12	0.421	17.63
Risk taking	0.140	4.14	−0.457	−22.31
Patience (Q43)	0.049	3.63	−0.069	−19.75
Positive reciprocity	0.124	3.02	0.103	5.66
Summary of entanglement	Trained adapters		Sign-profile prompt	
Own effects against the intended direction	0 of 7		3 of 7	
Bits dominant on their own surface (V5)	2 of 6		2 of 6	
Same, on standardized effects	2 of 6		1 of 6	
Surfaces whose largest mover is their own bit	1 of 7		1 of 7	
Surfaces whose largest mover is the trust bit	3 of 7		7 of 7	
Variance on the first principal component	51.3%		83.5%	
Mean absolute correlation between surfaces	0.395		0.807	

Note: Main effects are differences in mean surface composite between profiles with the bit set and without it, from an OLS fit with the six bits and their 15 two-way interactions (42 error degrees of freedom); composites are expected recoded responses averaged over the surface's items, so the units differ across surfaces. The manipulation is of synthetic training labels or prompted signs; the effects are changes in model response distributions. V5 was a planned acceptance check (at least 4 of 6 bits); the other summary rows are exploratory.

S8 Retraining noise by surface

The article bounds the strict-surface shifts of the unseeded retrain by their maximum. At the level of scored cells on the strict battery (140 cells), the largest single-cell difference in code probability is 0.052 and the mean absolute difference is 0.0060, and the Pearson correlation of expected codes over the 23 items is 0.9998. On the held-out pool (1087 cells) the corresponding values are 0.092, 0.0056 and 0.9996 over 223 items. Table S8 gives the shift on each surface, in standard deviations of that surface over the 64 profiles. These shifts are the noise floor that the near-zero reading in Section S9 uses. On the recoded scale of the readout, the Pearson correlation between the retrain and the original over the 23 strict items is 0.9999.

Table S8: Unseeded retrain minus original, strict-battery surfaces, in profile standard deviations (R, model side only).

Surface	Shift
Trust	−0.025
Risk taking	−0.030
Positive reciprocity	−0.011
Altruism	−0.008
Negative reciprocity	0.033
Patience (Q43)	0.006
Patience (Q50)	−0.112

S9 The near-zero margin in full

The near-zero check refits two predeclared countries without the training pairs of every dimension whose GPS coordinate lies within 0.10 of zero, each beside a size-matched control trained on the same number of pairs drawn at random. For Great Britain (profile `P_1_1_1_1_0_1`) the rule removed altruism, negative reciprocity, positive reciprocity and risk taking, so trust and the two patience items stayed in training. For Colombia (profile `P_1_0_0_1_1_0`) it removed altruism, risk taking and trust. Table S9 gives every contrast. The predeclared reading uses only the retained imprint surfaces, which are Great Britain's trust and Colombia's positive reciprocity. The dropped dimensions are a loss-of-signal check: with no training pairs left on a dimension, its surface is expected to move away from its declared sign, so a move there bears on neither side.

The floor in the last column is the unseeded retrain's shift on the same surface (Table S8), and almost every contrast exceeds it. That floor comes from one retrain on the full training data, and it is too small for refits trained on a fraction of the data.

The size-matched controls show why: a control differs from the full-data baseline by up to 1.222 standard deviations for Great Britain and 1.499 for Colombia. Against that variation, neither retained imprint contrast stands out. We therefore claim no reversal, and the margin stays a footnote in the article.

Table S9: Near-zero refit minus its size-matched control, in profile standard deviations (N, model side only).

Country	Surface	In training	Intended	Refit − control	Floor
Great Britain	Trust*	retained	up	0.448	0.025
	Patience (Q43)	retained	up	0.078	0.006
	Patience (Q50)	retained	up	0.389	0.112
	Risk taking	dropped	up	−0.018	0.030
	Positive reciprocity	dropped	down	0.219	0.011
	Negative reciprocity	dropped	up	−0.751	0.033
	Altruism	dropped	up	0.242	0.008
Colombia	Positive reciprocity*	retained	up	−0.092	0.011
	Negative reciprocity	retained	down	−0.931	0.033
	Patience (Q43)	retained	down	−0.223	0.006
	Patience (Q50)	retained	down	−0.559	0.112
	Trust	dropped	up	−0.480	0.025
	Risk taking	dropped	down	0.086	0.030
	Altruism	dropped	up	0.062	0.008

Note: Rows marked * are the retained imprint surfaces that the predeclared reading uses. "Intended" is the direction of the unit's own sign bit. Composites are frozen-recode item means, scaled by the standard deviation of the 64 second-cohort profile composites on the same surface. The floor is the unseeded retrain's absolute shift on that surface.

S10 Items selected by the probe

Table S10 lists the 56 items the probe selected under the primary rule, by assigned dimension and sign. A plus sign means that profiles with the dimension's bit set have a higher expected response code on the item. A minus sign means a lower one. Codes follow the WVS Wave 7 questionnaire, so the substantive meaning of a direction depends on each item's response scale. The list is model-side output, frozen before any held-out human value was read. As the article reports, item-level assignments are more stable under readout variants than membership in this particular list.

Table S10: The frozen primary selection: WVS Wave 7 items by assigned dimension and sign.

Dimension	Items	Sign +	Sign −
Trust	12	Q49	Q45, Q118, Q120, Q130, Q170, Q197, Q224, Q228, Q229, Q236, Q253
Risk taking	13	Q97, Q102, Q163, Q194	Q5, Q79, Q136, Q141, Q151, Q166, Q168, Q212, Q215
Positive reciprocity	11	Q95, Q105	Q6, Q15, Q17, Q51, Q52, Q53, Q54, Q55, Q205
Altruism	20	Q96, Q122, Q125, Q135, Q138, Q198, Q227, Q231, Q241	Q27, Q124, Q128, Q150, Q160, Q161, Q162, Q209, Q213, Q216, Q242

Note: The selection file's SHA-256 hash is in Section S12. Amendment 1 repaired the wording of 12 items after a first probe pass had been computed and read, and two of them, Q153 and Q157, were in that pass's top quartile by $|t|$. Neither is in this selection, which was made on the repaired wording. Q253 is the only repaired item selected.

S11 Descriptive follow-up analyses in full

This section gives the detail behind the four descriptive follow-up analyses of the article (surface D). None of them was preregistered, none enters a registered family, and none earns a claim. All of them use the strict battery and the 42 countries, and every comparison with respondents is retrospective, because the WVS Wave 7 file is the only human source and existed before the study. Human answer shares and country-level human values are used to compute the statistics and are not reported.

First-cohort adapters against the GPS and the human composite (D). Table S11 gives the three Spearman correlations of the article's three-link figure for every surface: first-cohort adapter composite with the GPS, adapter composite with the human composite of the same items, and the human composite with the GPS. All seven adapter–human intervals include zero, which is not evidence that no association exists, and the human–GPS interval excludes zero on four of the seven surfaces.

Second-cohort adapters against the GPS and the human composite (D). Table S12 repeats the first two columns of Table S11 for the second-cohort adapters, with each country assigned its profile's adapter. The human–GPS column does not depend on the adapters and is in Table S11. Positive reciprocity is the only adapter–human interval that excludes zero, at 0.366 [0.007, 0.636], and its lower limit is close to zero.

Table S11: Agreement with the GPS and with the WVS Wave 7 composite, first-cohort adapters, strict battery (surface D, descriptive and not preregistered; retrospective). Spearman correlation over countries, by surface.

Surface	Adapter–GPS	Adapter–human	Human–GPS
Trust	0.744 [0.500, 0.865]	0.295 [−0.044, 0.547]	0.351 [0.088, 0.564]
Risk taking	0.435 [0.085, 0.686]	0.120 [−0.260, 0.426]	−0.059 [−0.429, 0.312]
Positive reciprocity	0.488 [0.120, 0.695]	0.324 [−0.074, 0.619]	0.054 [−0.269, 0.386]
Altruism	0.406 [0.020, 0.676]	0.128 [−0.170, 0.369]	0.031 [−0.306, 0.350]
Negative reciprocity	−0.028 [−0.418, 0.406]	0.389 [−0.008, 0.694]	0.416 [0.173, 0.586]
Patience (Q43)	−0.019 [−0.414, 0.452]	0.192 [−0.169, 0.464]	−0.533 [−0.741, −0.266]
Patience (Q50)	0.177 [−0.313, 0.580]	0.003 [−0.291, 0.316]	0.388 [0.095, 0.628]

Note: Each cell is the Spearman correlation with a 95% interval. Over 42 countries, except the two human columns of risk taking, which cover 41 countries because one country has no valid answer to the risk item (Q178). Adapter intervals resample the 26 sign profiles (2,000 draws, seed 20261011); the human–GPS interval resamples countries (seed 20260905). The human composite is the Wave 7 item-mean composite of the same strict-battery items. The adapter–GPS column repeats the S1 values of the article.

Table S12: Second-cohort adapters against the GPS and the WVS Wave 7 composite, strict battery (D, descriptive; retrospective). Spearman correlation over countries.

Surface	Adapter–GPS	Adapter–human
Trust	0.803 [0.615, 0.876]	0.340 [−0.025, 0.586]
Risk taking	0.210 [−0.199, 0.584]	0.213 [−0.172, 0.533]
Positive reciprocity	0.366 [−0.068, 0.652]	0.366 [0.007, 0.636]
Altruism	0.364 [−0.057, 0.653]	0.094 [−0.181, 0.331]
Negative reciprocity	0.239 [−0.237, 0.604]	0.167 [−0.219, 0.522]
Patience (Q43)	0.005 [−0.403, 0.470]	0.218 [−0.121, 0.490]
Patience (Q50)	0.161 [−0.343, 0.557]	0.037 [−0.247, 0.332]

Note: Profile-block intervals over the 26 sign profiles (2,000 draws, seed 20261011). The risk-taking adapter–human column covers 41 countries, because one country has no valid answer to the risk item.

Distance to the survey distributions (D). Table S13 gives the four cells, Table S14 gives the mean total variation distance by GPS dimension, and Table S15 gives every paired contrast with both intervals. The dimension columns are point estimates, country-equal within the dimension's items, with no interval. The pooled column averages over the country–item cells instead of countries, and it differs from the country-equal mean only through the one cell that has no valid human answer. The persona prompt is closer to the survey than the base model on every dimension. The adapters are farther than the base model on trust, risk taking, positive reciprocity and negative reciprocity, and closer on patience and altruism. Adding the persona prompt to the adapters moves the distance by −0.096 [−0.115, −0.078], and adding the adapters to the persona prompt moves it by 0.055 [0.016, 0.097].

Table S13: Distance to the survey answer distributions in four cells (D, descriptive and not preregistered; retrospective). Lower is closer.

Cell	Mean TVD	Minus base (profile blocks)	Minus base (countries)	Trust–GPS
Base model	0.470	–	–	undefined
Country adapters	0.515	0.045 [0.006, 0.093]	[0.017, 0.079]	0.744 [0.500, 0.865]
Country-named persona prompt	0.365	−0.105 [−0.122, −0.086]	[−0.119, −0.091]	0.028 [−0.339, 0.381]
Adapters and persona prompt	0.420	−0.051 [−0.095, 0.005]	[−0.084, −0.012]	0.725 [0.505, 0.820]

Note: Mean total variation distance (TVD) between the model's probabilities over the answer codes and the survey-weighted shares of valid human answers, averaged over items within a country and then over the 42 countries. The means cover 965 country–item cells: the 23 items of the strict battery in each country, less one cell without a valid human answer. Contrasts are paired differences of country means, negative when the cell is closer to the survey than the base model; both intervals use 2,000 draws (profile blocks: seed 20261011; countries: seed 20260905). Trust–GPS is the Spearman correlation of the trust composite with the GPS over the countries (profile-block interval); it is undefined for the base model, whose trust composite is the same for every country.

The decomposition of the main effects (D). Table S16 gives the own-bit, full and cross-bit correlations with the GPS and the variance ratios for every surface. Table S17 gives, for each surface, the share of the variance over the 64 sign profiles that the six main effects carry, the correlation with the GPS of the prediction that adds two-way terms (where the main effects carry less than four fifths of the variance), and the rank correlation of the main-effects prediction with the composites that the second-cohort and first-cohort adapters give at the countries' sign profiles. The

Table S14: Mean total variation distance to the survey answer shares, by cell and GPS dimension (D, descriptive; retrospective). Lower is closer.

Cell	Trust	Patience	Risk	Pos. recip.	Neg. recip.	Altruism	All (countries)	All (cells)
Base model	0.413	0.664	0.565	0.384	0.411	0.494	0.470	0.470
Country adapters	0.479	0.655	0.614	0.436	0.508	0.439	0.515	0.515
Country-named persona prompt	0.315	0.541	0.524	0.244	0.352	0.244	0.365	0.365
Adapters and persona prompt	0.375	0.558	0.559	0.341	0.418	0.304	0.420	0.419

Note: Patience covers both patience items. The last two columns average over all items, by countries and pooled over the 965 country–item cells.

Table S15: Paired contrasts of mean total variation distance (D, descriptive; retrospective). Negative means that the first cell is closer to the survey.

Contrast	Profile-block interval	Country interval
Adapters − base	0.045 [0.006, 0.093]	[0.017, 0.079]
Persona − base	−0.105 [−0.122, −0.086]	[−0.119, −0.091]
Adapters and persona − base	−0.051 [−0.095, 0.005]	[−0.084, −0.012]
Adapters and persona − adapters	−0.096 [−0.115, −0.078]	[−0.111, −0.081]
Adapters and persona − persona	0.055 [0.016, 0.097]	[0.027, 0.086]
Interaction	0.009 [−0.003, 0.022]	[−0.001, 0.019]

Note: Differences of country means over 42 countries with 2,000 draws (profile blocks: seed 20261011; countries: seed 20260905). The interaction is the adapters' effect with the persona prompt minus the adapters' effect without it. The point estimate is the same under both resampling schemes.

main-effects prediction orders the second-cohort composites well on every surface, from 0.748 for risk taking to 0.957 for altruism. It orders the first-cohort composites less well on risk taking, 0.399, and negative reciprocity, 0.477, where the second-cohort effects transfer worst. The loss of the GPS order is therefore not a poor fit of the decomposition to the second-cohort adapters. It describes what the bits do.

Table S16: Own-bit and cross-bit parts of the second-cohort adapters' main effects, evaluated at each country's sign profile (D, descriptive and not preregistered). Spearman correlation with the GPS over the countries, and the cross part's share of the variance of the full prediction.

	Spearman with the GPS			Cross part of the full prediction	
Surface	Own bit only	Full	Cross bits only	Var(cross) / Var(full)	Covariance share
Trust	0.862	0.834	0.574	0.147	0.287
Risk taking	0.862	0.104	−0.097	0.999	0.949
Positive reciprocity	0.866	0.378	0.262	0.854	0.911
Altruism	0.857	0.422	0.285	0.850	0.911
Negative reciprocity	0.866	0.386	−0.128	0.830	0.727
Patience (Q43)	0.817	−0.020	−0.195	1.049	0.985
Patience (Q50)	0.817	0.176	0.009	0.914	0.935

Note: For each surface, the full prediction at a country's sign profile is the sum over the six bits of the bit's main effect on the surface times the bit's code (0.5 in absolute value); the own part keeps the surface's own bit and the cross part the other five. Evaluated at the sign profiles of the 42 countries and correlated with the GPS coordinate of the surface's own dimension. Var(cross) / Var(full) is a ratio of variances and not a share of variance explained, and it exceeds one when the own and cross parts covary negatively; the covariance share is Cov(cross, full) / Var(full). Model quantities and published GPS values only.

Held-out label recovery with both intervals (D). Table S18 gives the four recovery quantities with the profile-block and the country interval. The two intervals agree closely on every quantity, and the base model's accuracy interval includes one half under both.

S12 Provenance of the scoring runs

Specifications. Each scoring condition has a pinned specification hash that fixes the prompt template, the option order, the leading space before each code and the recode. Every run reproduced its pin before a model was loaded. The strict battery's pin begins `ed3bb964b606`, and the amended held-out pool's pin begins `46447de63f7c`. The pre-amendment held-out pin, which Amendment 1 superseded, began `4d799b02`.

Table S17: Fit of the main-effects decomposition, by surface (D, descriptive). Spearman correlations over the 42 countries unless stated.

Surface	Main-effect share of variance	Main + two-way with GPS	Full vs second-cohort composite	Full vs first-cohort composite
Trust	0.806	–	0.920	0.915
Risk taking	0.609	0.331	0.748	0.399
Positive reciprocity	0.711	0.287	0.935	0.934
Altruism	0.925	–	0.957	0.975
Negative reciprocity	0.666	0.210	0.835	0.477
Patience (Q43)	0.483	−0.066	0.871	0.826
Patience (Q50)	0.891	–	0.948	0.946

Note: The main-effect share is over the 64 sign profiles. Two-way terms are reported only on the four surfaces where the main effects carry less than four fifths of the variance. Model quantities and published GPS values only.

Table S18: Held-out synthetic-label recovery of the first-cohort adapters (D, descriptive; a construction check).

Quantity	Profile-block interval	Country interval
Recovery (adapter margin above base margin)	0.940 [0.926, 0.952]	[0.930, 0.949]
Accuracy (adapter prefers its own chosen completion)	0.816 [0.771, 0.856]	[0.786, 0.848]
Base accuracy	0.489 [0.471, 0.506]	[0.476, 0.501]
Mean margin shift (nats)	40.3 [28.2, 54.1]	[32.6, 49.5]

Note: Country-equal means over the 42 adapters and the held-out pairs of the training split, which the adapters never saw. The margin is the log probability of the chosen completion minus that of the rejected one. Recovery is the share of held-out pairs on which the adapter's margin exceeds the base model's. The intervals use 2,000 draws (profile blocks: seed 20261011; countries: seed 20260905).

Code revisions. All manifests of the held-out runs (the primary readout, the three readout variants and the prompted profiles) record the checkout revision `2126045`. During a recovery, three launcher files around that checkout were patched in place on the virtual machine to fix how a child environment was built. The scoring modules are byte-identical between the patched setup and the recorded revision, which the command `git diff -quiet 2126045 HEAD - paper_b/score_runtime.py paper_b/score_plan.py paper_b/score_cohort.py` confirmed at the time. The strict-battery run of the second cohort spans two recorded revisions. All but three of its manifests record `5133387`, and the remaining three (profiles `P_0_1_1_1_1_1`, `P_1_1_1_1_1_0` and `P_1_1_1_1_1_1`) record `f34a79f`. The unseeded retrain and the strict scoring of the prompted profiles ran at `b2dbbfb`, and the near-zero refits at `d1ab226`. Between these revisions the readout path changed in four ways, none of which touches prompt construction, option parsing, the forward pass or the frozen readout constants. A unit-name resolution replaced upper-casing, and it is the identity for every country and profile unit. A check on the number of training pairs reads a per-unit expectation that equals 526 for every unit outside the near-zero refits. Some functions gained annotations and parameters with backward-compatible defaults. The scoring loop gained a guard that stops with a named error when a scoring function is missing. The readout path is therefore not byte-identical across revisions, and what is verified is that every pin reproduces at the current revision and that the changes leave every scored unit's arithmetic unchanged.

The first-cohort scoring run. The first cohort's scores (surface S1) were persisted by a run whose manifest records a digest of the code bundle rather than a git commit, and none of the tracked source snapshots matches that digest. The commit that produced the run therefore cannot be named. Two properties of the run were verified directly. Its specification hash, which fixes the prompts, equals the strict pin, and the weights of the one adapter checked match the archived manifest byte for byte. A later replay of five selected items on the base model and that adapter, using the same four-bit quantized base model, lands within three units of half-precision rounding of the stored option log probabilities, which is the precision at which they were stored. The same replay on the unquantized model departs by far more. Quantization is therefore the source of the dominant difference between the two configurations, and the stored first-cohort scores are consistent with the quantized model that the adapters were trained on. The full panel was not replayed.

The archive of the development stage. Before every scoring run, a gate compared the base model's scores on the frozen software stack with the archived scores of the development stage, whose software environment was not pinned. The comparison is report-only by design. The formatted prompts matched the archive exactly, while the log probabilities differed by more than the tolerance used at the development stage. A replay with the same four-bit quantization moved the scores toward the archive

without reproducing it, so the remaining cause of that gap is not established. The gap enters no statistic in the article, which uses only scores produced on the frozen stack.

The follow-up runs. The adapters-plus-persona cell was scored in one run of the first-cohort adapters with the country-named persona prompt on the strict battery (run `adapter-persona-strict-v1-20261019`, pinned specification `persona_strict_v1`). Three gates passed before any country was scored: the fresh base model reproduced the banked strict base scores, one adapter without the persona prompt reproduced its banked first-cohort rows and differed from the base, and disabling the adapter returned the base. The held-out label recovery was scored in one run of the same adapters (run `label-recovery-cohort1-v1-20261019`), with the recipe that built the reference cache of the training pairs. Two gates passed before any country was scored: the base model reproduced the banked reference log probabilities on training pairs with a gap of zero, and one adapter changed the log probabilities while disabling it returned the base exactly. All country files were read back from the artifact store by hash. The code of both runs was written and run by the study team, and an independent audit of the two runs is pending.

Configuration. The configuration file `paper_b/config.py` serves both cohorts. Its first-cohort defaults are unchanged when the environment variable `SCA2_PB_UNITS` is unset, and the file is byte-identical to its version at the registration tag.

File hashes. Hashes of text files are given for the working-tree bytes. The item pool `pool_skeleton.csv` is written with Windows line endings by the pool-building script, while the repository stores text with Unix line endings, so the hash of the stored file differs from the hash of the file on disk. Both are given below. A reader who checks the freeze guard on a fresh checkout must regenerate the pool with `python paper_b/heldout/build_pool.py`, which reproduces the hashed bytes from the overrides file, because the guard compares raw bytes and refuses to run on the converted copy.

- Amended item pool, as written:
  `19241f6f103c51d64e295045162bb370920223b7ffad7c5f3c8991f5c101bd44`
- Amended item pool, as stored in the repository:
  `a1dbceea1a33da05f0432e14e0ff174ba239916b1a5beb9944e9aa0910cd40a5`
- Overrides file:
  `0f5cd2bbd88ef6eaca9cca9ded24e8a5e056276be055b21c1d92b941dd7d3849`
- Frozen primary and secondary selection:
  `782b7d2123f9758ea5641c56ddac6bf778fa4a01e3b6e7ce11402d224480bf21`

S13 Design detail and robustness numbers that the article summarizes

The item pool. The pool was built from WVS Wave 7 variable labels, value labels and questionnaire text alone. An item enters if it is a substantive question outside the battery used to develop the adapters, below the demographic block, with two to eleven contiguous ordinal response codes. A text review excluded items whose code order is not monotone or that test factual knowledge, confirmed the ordinal reading of others, and repaired wording that did not extract from the questionnaire, recording a reason for every change. The result is 223 items that no adapter was built for.

The two questionnaire readers. The face-valid rule assigns an item to a dimension when exactly one dimension's construct words occur in its wording and reads the direction from the response labels; a person could only confirm, flip or abstain, with a written reason. The typed decision model returns calibrated probabilities, given the item wording, the response options and the six GPS construct definitions used to generate the training data. Because single calls were unstable near the decision threshold, the comparator of record is 5 independent runs, and an item is assigned only when at least 3 runs agree on dimension and sign. Neither reader saw any model score or human response.

The wording repairs of Amendment 1. The 12 repaired items had sentences cut off mid-clause, a stripped country placeholder, grid rows truncated to a pronoun, or follow-ups that refer to a question absent from a standalone prompt. Each was repaired from its own response options or variable labels, under the pool rule and override mechanism already registered.

The country-named prompt run. The persona run first rescored the base model, and those scores reproduced the banked base-model scores of the second cohort's run exactly, so the persona column and the adapter columns share one software stack. The first-cohort values of the second-bank check reproduce the banked run exactly.

The primary screen: positive items and unfielded cells (H). Of the selected items, 60.7% have a positive item agreement. The human data contain 174 item–country cells that were never fielded; they enter as missing.

The registered reliability checks in full (H). The secondary rule keeps every imprint item above the fixed threshold. It gives $A = 0.070$ over 185 items, with a null ninety-fifth percentile of 0.034. Under the secondary rule, risk taking alone has an interval above zero, at 0.143 [0.045, 0.229] over 50 items, and its Bonferroni interval stays above zero, [0.017, 0.253]. The exploratory comparison on the 34 items fielded

in both waves gives $A = 0.043$ against wave-6 country means and 0.047 against wave-7 means over the 28 countries with both waves, so the small signal does not depend on one wave.

Readout variants: item membership (H). The correlation of $|t|$ over the pool between the original readout and each variant is 0.176 for reversed options, 0.546 for the first paraphrase and 0.271 for the second. The membership counts are in the article. The primary rule is a top-quartile cut on $|t|$, so small shifts reorder the items near the cut.

The cross-talk ratio across surfaces (D). Across the seven surfaces, the rank correlation between Var(cross)/Var(full) and the adapters' failure to order countries like the GPS is 0.821 against second-cohort composites and 0.286 against first-cohort composites. The comparison rests on trust set against six other surfaces and is not a graded relation; we claim no dose–response.

Cross effects between bits and surfaces in the factorial (F). The article gives two examples of a bit that moves another surface by more than its own. Further cross effects on the trained route: the risk-taking bit moves the negative-reciprocity surface by -0.668, and the trust bit moves patience Q50 by 0.495 and positive reciprocity by 0.418. Under prompting, the trust bit alone moves risk taking by -0.858 and patience Q50 by 1.774. The own effects are in Table S7.

Dropped dimensions in the near-zero refits (N). On the dimensions with no training pairs, 4 of 7 surfaces moved against their intended direction beyond the declared floor, as expected when a dimension has no training pairs. This is a loss-of-signal check and bears on neither side.